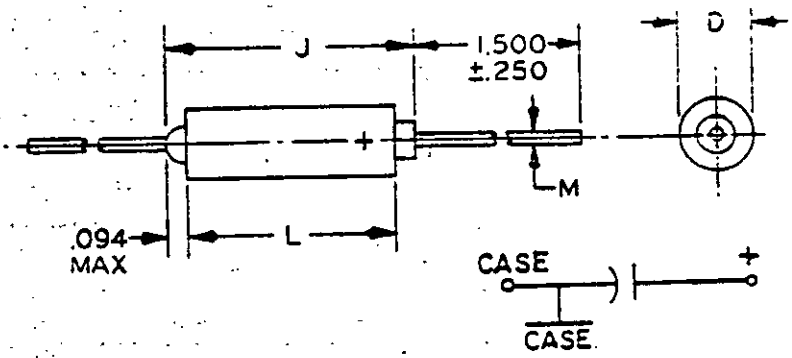
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), ESTABLISHED RELIABILILTY, STYLE CSR33

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39003.

FAILURE RATE LEVELS "M", "P" AND "R" ARE INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1984.



CIRCUIT DIAGRAM

Case size	Dimensions L ±.031	D +.016 -.015	M ±.002	M max
A	.286	.135	.020	.422
B	.474	.185	.020	.610
C	.686	.289	.025	.822
D	.786	.351	.025	.922

INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.135	3.43	.474	12.04
.015	.38	.185	4.70	.610	15.49
.016	.41	.250	6.35	.686	17.42
.020	.51	.286	7.26	.786	19.96
.025	.64	.289	7.34	.822	20.88
.031	.79	.351	8.92	.922	23.42
.094	2.39	.422	10.72	1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The case insulation shall extend .015 (.38 mm) minimum beyond each end. However, when a shrink-fitted insulation is used, it shall lap over the ends of the capacitor body.
Ⓐ 4. Lead length may be a minimum of 1-inch long for use in tape and reel automatic insertion equipment, when specified.

FIGURE 1. Dimensions and configurations.

FSC 5910

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tubular, insulated.

Material - Metal (alloy unspecified).

Type of seal - Hermetic.

Leads: Types N-4, N-5, W-4, or W-5 of MIL-STD-1276.

DC rated voltage: See table I. Above 85°C voltage derating is required (see MIL-C-39003).

DC leakage (DCL): See table I.

Capacitance (Cap.)(nom): See table I.

Cap. tolerance: See table I.

Dissipation factor (DF): See table I.

Operating temperature: -55°C to +125°C.

Failure rate level: See table I.

Stability at low and high temperatures: In accordance with MIL-C-39003.

Step 1 (+25°C):

DCL - See table I.

Cap. - Within tolerance of table I value.

DF - See table I.

Step 2 (-55°C):

Cap. - Shall not change more than ±10 percent of step 1 measured value.

DF - See table I.

Step 3 (+25°C):

DCL - See table I.

Cap. - Shall not change more than ±2 percent of step 1 measured value.

DF - See table I.

Step 4 (+85°C):

DCL - See table I.

Cap. - Shall not change more than ±8 percent of step 1 measured value.

DF - See table I.

Step 5 (+125°C):

DCL - See table I.

Cap. - Shall not change more than ±12 percent of step 1 measured value.

DF - See table I.

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Step 6 (+25°C):

DCL - See table I.

Cap. - Shall not change more than ±2 percent of step 1 measured value.

DF - See table I.

Surge voltage: In accordance with MIL-C-39003.

DCL - See table I.

Cap. - Shall not change more than ±2 percent of initial measured value.

DF - See table I.

Surge current 1/: Capacitors shall be subjected to 3 consecutive cycles of the conditions specified as follows;

Temperature - 25°C ±5°C.

Apply rated voltage ±2 percent for 4 ±1 seconds.

Discharge the capacitors for 4 ±1 seconds to a voltage below 1 percent of rated voltage.

The total dc resistance (excluding the capacitor) including the wiring, fixturing and output impedance of the regulated power supply to each test position during the charging cycle shall be 1.0 ±0.2 ohms.

Upon completion of the test, the capacitors shall meet the following requirements:

DCL - See table I.

Cap. - Shall not change more than ±2 percent of initial measured value.

DF - See table I.

Life: Method 108 of MIL-STD-202.

240, 1,000, and 2,000 hours:

At 25°C (as applicable):

DCL - See table I.

Cap. - Shall not change more than ±2 percent of initial measured value.

DF - See table I.

At 85°C:

DCL - See table I.

At 125°C:

DCL - See table I.

1/ This test shall be performed following thermal shock and voltage aging on initial qualification and on each inspection lot (group A) thereafter. A 5 percent defective allowed shall apply for group A and FRL's P, R, and S.

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Ⓐ TABLE I. Electrical characteristics and part numbers.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at			Dissipation factor at		Case size	Part no. M39003/03- Failure rate level for (%1,000 hrs)						
			+25°C	+85°C	+125°C	-55°C +25°C	+85°C +125°C		M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	µA	percent	percent								
6	10.0	10	0.5	2.0	2.0	6	6	A	0001	0101	0201	0301	2001	3001	4001
6	10.0	20	0.5	2.0	2.0	6	6	A	0002	0102	0202	0302	2002	3002	4002
6	12.0	10	0.5	2.0	2.0	6	6	A	0003	0103	0203	0303	2003	3003	4003
6	100.0	10	1.0	3.0	3.0	8	8	B	0004	0104	0204	0304	2004	3004	4004
6	100.0	20	1.0	3.0	3.0	8	8	B	0005	0105	0205	0305	2005	3005	4005
6	330.0	10	2.0	8.0	8.0	8	8	C	0006	0106	0206	0306	2006	3006	4006
6	330.0	20	2.0	8.0	8.0	8	8	C	0007	0107	0207	0307	2007	3007	4007
6	390.0	10	2.0	8.0	8.0	10	10	C	0008	0108	0208	0308	2008	3008	4008
6	470.0	10	2.0	8.0	8.0	10	10	C	0009	0109	0209	0309	2009	3009	4009
6	470.0	20	2.0	8.0	8.0	10	10	C	0010	0110	0210	0310	2010	3010	4010
6	680.0	10	5.0	10.0	10.0	10	10	D	0011	0111	0211	0311	2011	3011	4011
6	680.0	20	5.0	10.0	10.0	10	10	D	0012	0112	0212	0312	2012	3012	4012
6	820.0	10	5.0	10.0	10.0	10	10	D	0013	0113	0213	0313	2013	3013	4013
6	1000.0	10	5.0	10.0	10.0	10	10	D	0014	0114	0214	0314	2014	3014	4014
6	1000.0	20	5.0	10.0	10.0	10	10	D	0015	0115	0215	0315	2015	3015	4015
10	6.8	10	.5	2.0	2.0	6	6	A	0016	0116	0216	0316	2016	3016	4016
10	6.8	20	.5	2.0	2.0	6	6	A	0017	0117	0217	0317	2017	3017	4017
10	8.2	10	.5	2.0	2.0	6	6	A	0018	0118	0218	0318	2018	3018	4018
10	47.0	10	1.0	2.0	2.0	6	6	B	0019	0119	0219	0319	2019	3019	4019
10	47.0	20	1.0	2.0	2.0	6	6	B	0020	0120	0220	0320	2020	3020	4020
10	56.0	10	1.0	4.0	4.0	6	6	B	0021	0121	0221	0321	2021	3021	4021
10	68.0	10	1.0	4.0	4.0	6	6	B	0022	0122	0222	0322	2022	3022	4022
10	68.0	20	1.0	4.0	4.0	6	6	B	0023	0123	0223	0323	2023	3023	4023
10	82.0	10	1.0	4.0	4.0	6	6	B	0024	0124	0224	0324	2024	3024	4024
10	220.0	10	1.0	7.0	7.0	8	8	C	0025	0125	0225	0325	2025	3025	4025
10	220.0	20	1.0	7.0	7.0	8	8	C	0026	0126	0226	0326	2026	3026	4026
10	270.0	10	2.0	10.0	10.0	8	8	C	0027	0127	0227	0327	2027	3027	4027
10	390.0	10	2.0	16.0	16.0	10	10	D	0028	0128	0228	0328	2028	3028	4028
10	470.0	10	4.0	16.0	16.0	10	10	D	0029	0129	0229	0329	2029	3029	4029
10	470.0	20	4.0	16.0	16.0	10	10	D	0030	0130	0230	0330	2030	3030	4030
10	560.0	10	4.0	16.0	16.0	10	10	D	0031	0131	0231	0331	2031	3031	4031
15	4.7	10	.5	2.0	2.0	4	4	A	0032	0132	0232	0332	2032	3032	4032
15	4.7	20	.5	2.0	2.0	4	4	A	0033	0133	0233	0333	2033	3033	4033
15	5.6	10	.5	2.0	2.0	4	4	A	0034	0134	0234	0334	2034	3034	4034
15	33.0	10	1.0	2.0	2.0	6	6	B	0035	0135	0235	0335	2035	3035	4035
15	33.0	20	1.0	2.0	2.0	6	6	B	0036	0136	0236	0336	2036	3036	4036
15	39.0	10	1.0	2.0	2.0	6	6	B	0037	0137	0237	0337	2037	3037	4037
15	150.0	10	1.0	7.0	7.0	8	8	C	0038	0138	0238	0338	2038	3038	4038
15	150.0	20	1.0	7.0	7.0	8	8	C	0039	0139	0239	0339	2039	3039	4039
15	180.0	10	2.0	10.0	10.0	8	8	C	0040	0140	0240	0340	2040	3040	4040
15	220.0	10	2.0	10.0	10.0	8	8	D	0041	0141	0241	0341	2041	3041	4041
15	220.0	20	2.0	10.0	10.0	8	8	D	0042	0142	0242	0342	2042	3042	4042
15	270.0	10	2.0	16.0	16.0	8	8	D	0043	0143	0243	0343	2043	3043	4043
15	330.0	10	2.0	16.0	16.0	8	8	D	0044	0144	0244	0344	2044	3044	4044
15	330.0	20	2.0	16.0	16.0	8	8	D	0045	0145	0245	0345	2045	3045	4045

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Ⓐ TABLE I. Electrical characteristics and part numbers - Continued.

DC rated voltage	Capacitance	Capacitance tolerance	DC leakage at			Dissipation factor at		Case size	Part no. M39003/03- Failure rate level for (%1,000 hrs)						
			+25°C	+85°C	+125°C	-55°C +25°C	+85°C +125°C		M 1.0	P 0.1	R 0.01	S 0.001	B 0.1	C 0.01	D 0.001
volts	µF	percent	µA	µA	µA	percent	percent								
20	2.7	10	.5	2.0	2.0	4	4	A	0046	0146	0246	0346	2046	3046	4046
20	3.3	10	.5	2.0	2.0	4	4	A	0047	0147	0247	0347	2047	3047	4047
20	3.3	20	.5	2.0	2.0	4	4	A	0048	0148	0248	0348	2048	3048	4048
20	3.9	10	.5	2.0	2.0	4	4	A	0049	0149	0249	0349	2049	3049	4049
20	18.0	10	1.0	2.0	2.0	6	6	B	0050	0150	0250	0350	2050	3050	4050
20	22.0	10	1.0	2.0	2.0	6	6	B	0051	0151	0251	0351	2051	3051	4051
20	22.0	20	1.0	2.0	2.0	6	6	B	0052	0152	0252	0352	2052	3052	4052
20	27.0	10	1.0	2.0	2.0	6	6	B	0053	0153	0253	0353	2053	3053	4053
20	56.0	10	1.0	10.0	10.0	6	6	C	0054	0154	0254	0354	2054	3054	4054
20	68.0	10	1.0	10.0	10.0	6	6	C	0055	0155	0255	0355	2055	3055	4055
20	68.0	20	1.0	10.0	10.0	6	6	C	0056	0156	0256	0356	2056	3056	4056
20	82.0	10	1.0	10.0	10.0	6	6	C	0057	0157	0257	0357	2057	3057	4057
20	100.0	10	1.0	10.0	10.0	6	6	C	0058	0158	0258	0358	2058	3058	4058
20	100.0	20	1.0	10.0	10.0	6	6	C	0059	0159	0259	0359	2059	3059	4059
20	120.0	10	1.0	10.0	10.0	6	6	C	0060	0160	0260	0360	2060	3060	4060
20	150.0	10	2.0	10.0	10.0	8	8	D	0061	0161	0261	0361	2061	3061	4061
20	150.0	20	2.0	10.0	10.0	8	8	D	0062	0162	0262	0362	2062	3062	4062
20	180.0	10	2.0	10.0	10.0	8	8	D	0063	0163	0263	0363	2063	3063	4063
35	1.8	10	.5	2.0	2.0	4	4	A	0064	0164	0264	0364	2064	3064	4064
35	8.2	10	1.0	2.0	2.0	6	6	B	0065	0165	0265	0365	2065	3065	4065
35	10.0	10	1.0	2.0	2.0	6	6	B	0066	0166	0266	0366	2066	3066	4066
35	10.0	20	1.0	2.0	2.0	6	6	B	0067	0167	0267	0367	2067	3067	4067
35	33.0	10	1.0	5.0	5.0	6	6	C	0068	0168	0268	0368	2068	3068	4068
35	33.0	20	1.0	5.0	5.0	6	6	C	0069	0169	0269	0369	2069	3069	4069
35	39.0	10	1.0	5.0	5.0	6	6	C	0070	0170	0270	0370	2070	3070	4070
35	47.0	10	1.0	5.0	5.0	6	6	C	0071	0171	0271	0371	2071	3071	4071
35	47.0	20	1.0	5.0	5.0	6	6	C	0072	0172	0272	0372	2072	3072	4072
35	56.0	10	2.0	10.0	10.0	6	6	D	0073	0173	0273	0373	2073	3073	4073
35	68.0	10	2.0	10.0	10.0	6	6	D	0074	0174	0274	0374	2074	3074	4074
35	68.0	20	2.0	10.0	10.0	6	6	D	0075	0175	0275	0375	2075	3075	4075
50	1.2	10	.5	2.0	2.0	4	4	A	0076	0176	0276	0376	2076	3076	4076
50	1.5	10	.5	2.0	2.0	4	4	A	0077	0177	0277	0377	2077	3077	4077
50	1.5	20	.5	2.0	2.0	4	4	A	0078	0178	0278	0378	2078	3078	4078
50	5.6	10	1.0	2.0	2.0	4	4	B	0079	0179	0279	0379	2079	3079	4079
50	6.8	10	1.0	2.0	2.0	6	6	B	0080	0180	0280	0380	2080	3080	4080
50	6.8	20	1.0	2.0	2.0	6	6	B	0081	0181	0281	0381	2081	3081	4081
50	22.0	10	1.0	5.0	5.0	6	6	C	0082	0182	0282	0382	2082	3082	4082
50	22.0	20	1.0	5.0	5.0	6	6	C	0083	0183	0283	0383	2083	3083	4083
50	27.0	10	1.0	5.0	5.0	6	6	C	0084	0184	0284	0384	2084	3084	4084
50	33.0	10	1.0	9.0	9.0	6	6	D	0085	0185	0285	0385	2085	3085	4085
50	33.0	20	1.0	9.0	9.0	6	6	D	0086	0186	0286	0386	2086	3086	4086
50	39.0	10	1.0	9.0	9.0	6	6	D	0087	0187	0287	0387	2087	3087	4087

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4,000, 6,000, 8,000, and 10,000 hours:

At 25°C (as applicable):

DCL - 200 percent of initial limit.

Cap. - Shall not change more than ±10 percent of initial measured value.

DF - See table I.

At 85°C:

DCL - 200 percent of initial limit.

At 125°C:

DCL - 200 percent of initial limit.

Marking: In accordance with MIL-C-39003.

Part number: M39003/06-(dash number from table I).

(A) Substitutability data:

Substitutability for failure rate levels shall be as follows: In accordance with MIL-C-39003.

Substitutability for capacitance tolerance shall be as follows:

Cap tolerance	Will replace cap tolerance
10%	20%

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AV
Navy - SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - MC, AS, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-39003/6A	CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYT TANTALUM, (POLARIZED, SINTERED SLUG), ESTABLISHED RELIABILITY, STYLE CSR33

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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